SOLUTIONS: CONTROL ENGINEERING

1. Exercise

a) By Cayley-Hamilton theorem we know that:

$$A^n + \alpha_{n-1}A^{n-1} + \alpha_{n-2}A^{n-2} + \ldots + \alpha_2 A^2 + \alpha_1 A + \alpha_0 I = 0.$$

Since $\det(A) \neq 0$, then $p(0) = \alpha_0 \neq 0$. Multiplying by A^{-n} the above equation yields:

$$I + \alpha_{n-1}A^{-1} + \alpha_{n-2}A^{-2} + \ldots + \alpha_2 A^{-(n-2)} + \alpha_1 A^{-(n-1)} + \alpha_0 A^{-n} = 0.$$

Hence,

$$A^{-n} = -\frac{I + \alpha_{n-1}A^{-1} + \alpha_{n-2}A^{-2} + \ldots + \alpha_2 A^{-(n-2)} + \alpha_1 A^{-(n-1)}}{\alpha_0}.$$

[4]

b) We solve the three sub-questions below:

i) To find the equilibria we solve the system of equations:

$$\begin{cases} x_1(1-x_1) - x_1 x_2 &= 0 \\ -x_2 + 2x_1 x_2 &= 0 \end{cases}$$

To solve the second equation we only have two possibilities: $x_2 = 0$ or $x_1 = 1/2$. The first one yields, by substitution into the first one $x_1(1-x_1) = 0$, viz. either $x_1 = 0$ or $x_1 = 1$. The second one yields: $1/4 - x_2/2 = 0$, viz. $x_2 = 1/2$. Overall we found 3 equilibrium points in:

$$x_{e1} = [0,0]^T \quad x_{e2} = [1,0]^T \quad x_{e3} = [1/2, 1/2]^T$$

. [4]

ii) The Jacobian matrix associated to the nonlinear system is given as:

$$J(x) = \begin{bmatrix} 1 - 2x_1 - x_2 & -x_1 \\ 2x_2 & 2x_1 - 1 \end{bmatrix}.$$

Evaluating the Jacobian at the 3 equilibrium points yields the following linearized dynamics:

$$\dot{\tilde{x}} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \tilde{x} \qquad \text{around } x_{e1}$$

$$\dot{\tilde{x}} = \begin{bmatrix} -1 & -1 \\ 0 & 1 \end{bmatrix} \tilde{x} \qquad \text{around } x_{e2}$$

$$\dot{\tilde{x}} = \begin{bmatrix} -1/2 & -1/2 \\ 1 & 0 \end{bmatrix} \tilde{x} \qquad \text{around } x_{e3}$$

[3]

iii) Computing the spectra of the matrix A associated to the linearized dynamics around x_{e1} yields:

$$\lambda_1 = 1 \quad \lambda_2 = -1.$$

Hence, the principle of stability in the first approximation can be applied to conclude instability of the original equilibrium due to an eigenvalue with positive real part. Similarly, computing the spectra of the matrix A associated to the linearized dynamics around x_{e2} yields:

$$\lambda_1 = -1 \quad \lambda_2 = 1.$$

Hence, the principle of stability in the first approximation can be applied to conclude instability of the original equilibrium x_{e2} due to an eigenvalue with positive real part. For the equilibrium x_{e3} we see:

$$\det(\lambda I - A) = (\lambda + 1/2)\lambda + 1/2 = \lambda^2 + \lambda/2 + 1/2.$$

The roots of the characteristic polynomial have negative real part (due to the permanence in sign of its coefficients) hence the equilibrium x_{e3} is locally asymptotically stable. [5]

c) Computing the state of the system at time 1, as a function of α yields:

$$x(1) = [\alpha(1 - e^{-1}), \alpha]^T$$

This is seen considering the second component $x_2(t) = x_2(0) + \int_0^t u(t)dt$, while for the first component: $x_1(t) = e^{-t}x_1(0) + \int_0^t e^{-(t-s)}u(s)ds$. At time 2 we have:

$$x(2) = [\alpha(1 - e^{-1})e^{-1} + \beta(1 - e^{-1}), \alpha + \beta]^T.$$

In order to have $x_2(2) = 0$ we need $\beta = -\alpha$. Moreover:

$$\alpha(1 - e^{-1})e^{-1} - \alpha(1 - e^{-1}) = 3$$

yields:

$$\alpha = -\frac{3}{(1 - e^{-1})^2} \text{ and } \beta = \frac{3}{(1 - e^{-1})^2}.$$

[6]

The state $x = [3, 0]^T$ is not an equilibrium point, since the equation:

$$\begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u = \begin{bmatrix} -3 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u = 0$$

has no solution (with respect to unknown u; this is because $[-3, 0]^T$ is not in the image of $[1, 1]^T$). [2]

d) We first need to compute the matrix representation.

i) The system can be written in matrix representation according to $x^+ = Ax + Bu$, $y = Cx$ with:

$$A = \begin{bmatrix} 1 & \alpha \\ 0 & -1 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, C = [1, 1].$$

[2]

ii) Accordingly the Reachability matrix is given by:

$$\mathscr{R} = [B, AB] = \begin{bmatrix} 0 & \alpha \\ 1 & -1 \end{bmatrix}$$

Therefore, $\det(\mathscr{R}) = -\alpha$ which gives reachability if and only if $\alpha \neq 0$. [2]

iii) On the other hand

$$\mathscr{O} = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & \alpha - 1 \end{bmatrix}.$$

Computing its determinant we see:

$$\det(\mathscr{O}) = \alpha - 2.$$

This gives observability if and only if $\alpha \neq 2$. [2]

iv) The eigenvalues of A are $\pm 1 \neq 0$ regardless of α, therefore reachability holds if and only if controllability holds. [2]

e) The powers of A read:

$$A^2 = \begin{bmatrix} 0 & 0 & 1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}$$

$$A^3 = -I$$

$$A^4 = -A$$

$$A^5 = -A^2$$

$$A^6 = I$$

i) The observability matrix is therefore:

$$\mathscr{O} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

which is clearly full-rank (being the identity). [4]

ii) Moreover, for $x(0) = [a, b, c]^T$ we have:

$$y(0) = [1,0,0]x(0) = a$$

$$y(1) = [1,0,0]Ax(0) = [0,1,0]x(0) = b$$

$$y(2) = [1,0,0]A^2x(0) = [0,0,1]x(0) = c$$

$$y(3) = [1,0,0]A^3x(0) = -[1,0,0]x(0) = -a$$

$$y(4) = [1,0,0]A^4x(0) = -[1,0,0]Ax(0) = -b$$

$$y(5) = [1,0,0]A^5x(0) = -[1,0,0]A^2x(0) = -c.$$

Moreover, the sequence repeats periodically for subsequent times, since $x(6) = A^6x(0) = x(0)$. [4]

2. Exercise

a) Clearly, with the considered choice of state variables we have:

$$\dot{x}_1 = x_2, \qquad \dot{x}_3 = x_4$$

On the other hand, by the second order equations:

$$J_1\dot{x}_2 = -k(x_1 - x_3) - \beta x_2 + u$$

$$J_2\dot{x}_4 = -k(x_3 - x_1) - \beta x_4$$

Accordingly, we may write the system as:

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -k/J_1 & -\beta/J_1 & k/J_1 & 0 \\ 0 & 0 & 0 & 1 \\ k/J_2 & 0 & -k/J_2 & -\beta/J_2 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1/J_1 \\ 0 \\ 0 \end{bmatrix} u$$

$$y = [0,0,1,0]x$$

[8]

b) For parameters equal to 1, the matrix A reads:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & -1 & -1 \end{bmatrix}$$

Hence the controllability matrix can be computed as:

$$\mathscr{R} = [B, AB, A^2B, A^3B] = \begin{bmatrix} 0 & 1 & -1 & 0 \\ 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

It can be seen that $\mathscr{R}$ is block triangular, with both blocks on the diagonal of full rank, hence the system is controllable. [6]

c) The observability matrix is given by

$$\mathscr{O} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & -1 & -1 \\ -1 & 1 & 1 & 0 \end{bmatrix}$$

It can be seen that $\det(\mathscr{O}) = 1$, thus showing that the system is observable. [6]

d) Straightforward calculations show that the considered matrix is the desired inverse. [2]

We design an observer of the form:

$$\dot{\hat{x}} = A\hat{x} + L(C\hat{x} - y) + Bu.$$

The error system $e = \hat{x} - x$ fulfills:

$$\dot{e} = (A + LC)e,$$

where L can be found from Ackermann's formula according to:

$$L = -[0,0,0,1]\mathscr{O}^{-T} p_d(A^T)$$

where p_d is the desired closed-loop polynomial, i.e. $p_d(\lambda) = (\lambda+1)^4 = \lambda^4 + 4\lambda^3 + 6\lambda^2 + 4\lambda + 1$. This yields $L = [1,-1,-2,1]^T$. [8]

3. Exercise

a) To verify symmetry of $\eta(t)$ remark:

$$\eta(t)^T = [e^{A^T t} Q e^{At}]^T = [[e^{At}]^T Q^T [e^{A^T t}]^T = e^{A^T t} Q e^{At} = \eta(t)$$

where we exploited $Q = Q^T$ and $[e^{At}]^T = e^{A^T t}$. [3]

b) Clearly $\eta(0) = I_2 \cdot Q \cdot I_2 = Q$. Moreover, taking its derivative with respect to time yields:

$$\dot{\eta}(t) = \frac{d}{dt}[e^{A^T t} Q e^{At}] = [\frac{d}{dt} e^{A^T t}] Q e^{At} + e^{A^T t} Q [\frac{d}{dt} e^{At}]$$

$$= A^T [e^{A^T t} Q e^{At}] + [e^{A^T t} Q e^{At}] A = A^T \eta(t) + \eta(t) A.$$

[5]

c) To show symmetry it is enough to remark:

$$P^T = \int_0^{+\infty} \eta(t)^T \, dt = \int_0^{+\infty} \eta(t) \, dt = P.$$

To prove well-posedness, notice that by the assumption on the eigenvalues of A, there exists positive constants M and λ such that $\|e^{At}\| \leq M e^{-\lambda t}$. Hence,

$$\left\| \int_0^{+\infty} \eta(t) \, dt \right\| \leq \int_0^{+\infty} \|\eta(t)\| \, dt \leq \|Q\| \int_0^{+\infty} M^2 e^{-2\lambda t} \, dt < +\infty.$$

[5]

d) To show positive definiteness we take any $x_0 \in \mathbb{R}^n \neq 0$. Clearly:

$$x_0^T P x_0 = \int_0^{+\infty} x_0^T \eta(t) x_0 dt = \int_0^{+\infty} x(t)^T Q x(t) \, dt$$

where $x(t) = e^{At} x_0$ is the solution of the linear system from initial condition x_0. Notice that $x(t)^T Q x(t) > 0$ for all t such that $x(t) \neq 0$ (by positive definiteness of Q). Since $x_0 \neq 0$, $x(t) \neq 0$ for all t. In particular then the integrand $x(t)^T Q x(t) > 0$ for all t, and as a consequence $x_0^T P x_0 > 0$. [5]

e) To prove the identity we see:

$$A^T P + PA = \int_0^{+\infty} A^T \eta(t) + \eta(t) A \, dt = \int_0^{+\infty} \dot{\eta}(t) \, dt$$

$$= \eta(t)|_0^{+\infty} = \eta(\infty) - \eta(0) = -Q.$$

[5]

f) According to the previous construction $\int_0^{+\infty} \eta(t) \, dt$ is a solution of the equation $A^T P + PA = -Q$. This is well defined because A has negative eigenvalues (in -1 and -2 respectively). [2]
To solve the equation with respect to P we write:

$$\begin{bmatrix} -1 & 1 \\ 0 & -2 \end{bmatrix}^T P + P \begin{bmatrix} -1 & 1 \\ 0 & -2 \end{bmatrix} = -I_2$$

We choose P of the following form:

$$P = \begin{bmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{bmatrix}.$$

Accordingly we get the following linear system of equations:

$$\begin{cases} -2p_{11} &= -1 \\ p_{11} - 3p_{12} &= 0 \\ 2p_{12} - 4p_{22} &= -1 \end{cases}$$

Solving the system yields:

$$p_{11} = 1/2 \qquad p_{12} = p_{11}/3 = 1/6 \qquad p_{22} = 1/3.$$

[5]